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Pseudocode

① Design a Class Hierarchy

```
START
CLASS Person
 Name, Age
 END CLASS
CLASS Student INHERITS Person
   RegNo, Marks[5]
  METHOD, Average()
   Avg = (M1 + M2 + M3 + M4 + M5) / 5
   END METHOD
 END CLASS
CLASS ResearchStudent INHERITS
Student
   Research Area
   METHOD Display()
    PRINT Name, Age, RegNo,
   ResearchArea, Avg
   END METHOD
END CLASS
CREATE ResearchStudent
INPUT details
CALL Average()
CALL Display()
STOP
```

② Complete the Missing Pseudocode:

```
BEGIN
READ N
READ ARRAY A
For i ← 0 TO N-1
  For j ← i+1 TO N-1
    IF A[i] == A[j]
      DELETE A[j]
    END IF
  END FOR
END FOR
PRINT Modified Array
END.
```

③ Debug the Pseudocode

```
BEGIN
CLASS Area
  METHOD area(side)
    RETURN side * side
  END METHOD
  METHOD area(length, breadth)
    RETURN length * breadth
  END METHOD
END CLASS
END.
```

(4) Trace the Execution

```
BEGIN
sum ← 0
For i ← 1 TO 5
  IF i MOD 2 == 0
    sum ← sum + i
  ELSE
    sum ← sum - i
  ENDIF
ENDFOR
PRINT sum
END.
```

(5) Write Efficient pseudocode:

```
BEGIN
READ N
READ Array A
first ← -∞
second ← -∞
For i ← 0 TO N-1
  IF A[i] > first THEN
    second ← first
    first ← A[i]
  ELSE IF A[i] > Second AND A[i] ≠
  first THEN
      second ← A[i]
    ENDIF
  ENDFOR
PRINT second
END.
```

6) Convert Problem statement into Pseudocode

```
BEGIN
INTERFACE Payment
  METHOD Pay(amount)
END INTERFACE
CLASS Credit Card IMPLEMENTS Payment
  Pay(amount)
END CLASS
CLASS UPI IMPLEMENTS Payment
  Pay(Amount)
END CLASS
CLASS NetBanking IMPLEMENTS Payment
  Pay(amount)
END CLASS
READ choice, amount
IF choice =1
  Payment ← credit card
ELSE IF choice =2
  Payment ← UPI
ELSE
  Payment ← Net Banking
ENDIF
CALL Payment.Pay(amount)
END.
```

(7) Complete the Pseudocode

```
BEGIN
CLASS Animal
METHOD speak()
PRINT "Animal"
END METHOD
END CLASS
CLASS Dog INHERITS Animal
METHOD speak()
PRINT "Barks"
END METHOD
END CLASS
Animal obj ← New Dog()
obj.speak()
END.
```

(8) Find Logical Errors

```
BEGIN
CLASS Employee
DATA
   id
   name
CONSTRUCTOR(id,name)
this.id ← id
this.name ← name
END CONSTRUCTOR
END CLASS
END.
```

Case Study (Integrated Problem)

```
BEGIN
CLASS Vehicle
  VehicleNo, ownerName
   METHOD display()
  END CLASS

  CLASS Car INHERITS Vehicle
    fuel Type
  END CLASS
  CLASS Truck INHERITS Vehicle
    loadCapacity
   END CLASS
  READ N
   For i ← 1 TO N
     READ type
     If type = "car"
        obj ← NEW car ()
      ELSE
       obj ← NEW Truck()
      END IF
     READ vehicleNo, ownerName
      STORE obj
     ENDFOR
    FOR EACH obj
    CALL obj.display()
    ENDFOR

    END
```